Extending FedAWE to Federated Generative Learning under Client Unavailability

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Abstract

Federated learning systems often suffer from heterogeneous and non-stationary client unavailability, where clients participate irregularly due to network, device, or user-side constraints. The FedAWE algorithm addresses this issue through adaptive innovation echoing and implicit gossiping, showing strong performance for federated classification tasks. In this project, we first reproduce the main FedAWE setting and then extend the problem to federated generative learning using Variational Autoencoders (VAEs). Our experiments show that, unlike in classification, vanilla FedAWE becomes unstable for generative training under severe dropout because stale echoed updates distort the learned latent distribution. To address this limitation, we propose Quality Echo filtering and Selective Echo FedAvg-VAE, which reuse stale updates only when they are recent and directionally compatible. Results show that Selective Echo substantially reduces the instability of FedAWE while preserving generative diversity close to FedAvg.

1. Motivation

Federated learning (FL) enables multiple clients to collaboratively train a global model without directly sharing their local data. This is especially useful in privacy-sensitive domains where data is distributed across devices or institutions. However, a major challenge in practical FL is client unavailability. At any communication round, some clients may be offline because of unstable connectivity, limited computation, battery constraints, or user activity.

The original FedAWE paper studies this problem under heterogeneous and non-stationary client availability. In their setting, client availability probabilities may differ across clients and change over time. The paper shows that such unavailability can bias standard FedAvg toward frequently available clients and proposes FedAWE to compensate for missed updates.

Our project follows two goals. First, we reproduce the original FedAWE behavior under client unavailability. Second, we extend the paper to a more challenging setting: federated generative learning. Instead of training only classifiers, we ask whether the FedAWE idea still works when the model must learn a full data distribution using a VAE. This extension is motivated by the fact that generative models are usually more sensitive to stale or inconsistent updates than discriminative classifiers.

Main Contributions

- We reproduce the original FedAWE classification setting under client unavailability.
- We extend FedAWE from federated classification to federated generative learning using VAEs.
- We show that stale replay destabilizes latent-space optimization under severe dropout.
- We propose Quality Echo and Selective Echo stabilization mechanisms.
- We validate the findings using MNIST, FashionMNIST, latent-space PCA analysis, and ablation studies.

2. Background

2.1. Federated Learning and FedAvg

In standard federated learning, a server coordinates m clients. Each client i has a local objective $F_i(x)$, and the global objective is

$$F(x) = \frac{1}{m}\sum_{i=1}^{m} F_i(x). \tag{1}$$

FedAvg trains this model by selecting available clients at each round, letting them perform local training, and then averaging their updated models. When client availability is uniform, this procedure can work well. However, when some clients are unavailable more often than others, the global model becomes biased toward the clients that participate more frequently.

2.2. FedAWE

FedAWE was proposed to reduce this bias under heterogeneous and non-stationary client unavailability. It introduces two main ideas.

First, **adaptive innovation echoing** compensates for missed local computation. When a client returns after being unavailable, its update is scaled based on the number of missed rounds. This allows the client to "catch up" without performing all missed local computations.

Second, **implicit gossiping** delays the broadcast of the global model until the end of the round. This causes participating clients to indirectly mix information through the server, without direct client-to-client communication.

The key assumption behind FedAWE is that stale or delayed local innovations still carry useful information for the global optimization process. This assumption is reasonable in the classification settings studied in the original paper. However, our project investigates whether the same assumption holds for generative models.

3. Reproduction of FedAWE

3.1. Reproduction Setup

For the reproduction part, we followed the original paper's federated classification setup. The original paper evaluates FedAWE on image classification datasets such as SVHN, CIFAR-10, and CINIC-10 using 100 clients and highly non-IID data generated using a Dirichlet distribution with $\alpha = 0.1$.

We reproduced the main training behavior using the FedAWE codebase and PyTorch. We tested FedAvg and FedAWE under client unavailability and verified that the training pipeline correctly simulates client dropout, local training, aggregation, and evaluation.

3.2. Comparison with Original Paper

The original paper reports that FedAWE consistently improves over FedAvg in classification tasks under different availability dynamics. For example, on CIFAR-10 under stationary unavailability, FedAWE reports a test accuracy of 66.3%, while FedAvg over active clients reports 62.9%. Similar improvements are shown under staircase, sine, and interleaved sine non-stationary dynamics.

Our reproduction confirmed the expected qualitative trend: FedAWE is designed to improve classification robustness under client unavailability. Small numerical discrepancies between our reproduction and the paper may arise from differences in hardware, random seeds, total training rounds, implementation details, and available computation budget.

Table 1. Representative comparison from the original FedAWE paper on CIFAR-10 test accuracy. FedAWE improves over FedAvg in the original classification setting.

Availability Setting	FedAvg	FedAWE
Stationary	62.9%	**66.3%**
Staircase	63.0%	**66.0%**
Sine	62.1%	**63.5%**
Interleaved Sine	60.7%	**63.3%**

4. Extension to Federated Generative Learning

4.1. Why Generative Models?

The original FedAWE paper focuses on classification. However, generative learning is a different and more sensitive setting. A classifier mainly learns decision boundaries, while a generative model must learn the full data distribution. This means stale or misaligned updates may not only reduce accuracy, but can distort the learned latent representation and degrade sample quality.

To study this, we replaced the classifier with a Variational Autoencoder (VAE). The VAE learns an encoder $q_\phi(z|x)$ and decoder $p_\theta(x|z)$, where z is a latent variable. The objective consists of a reconstruction term and a KL regularization term:

$$\mathcal{L}_{VAE} = \mathcal{L}_{recon}(x, \hat{x}) + D_{KL}(q_\phi(z|x)||p(z)). \quad (2)$$

We use this setting to ask the central research question:

> Does FedAWE still help when moving from federated classification to federated generative learning?

4.2. Federated VAE Setup

We trained a fully-connected VAE on MNIST using 20 clients. The data was split using a Dirichlet distribution with $\alpha = 0.1$ to create non-IID client datasets. We focused mainly on a high-dropout setting, where only a small number of clients are available in each round.

We compared four methods:

- **FedAvg-VAE**: standard federated averaging using only fresh updates.
- **FedAWE-VAE**: direct adaptation of FedAWE echoing to VAE training.
- **Quality Echo FedAWE-VAE**: FedAWE with staleness and cosine-based echo filtering.
- **Selective Echo FedAvg-VAE**: FedAvg as the base method, with selective reuse of safe echoed updates.

5. Proposed Improvements

5.1. Quality Echo Filtering

Directly applying FedAWE to VAEs caused instability. The reason is that FedAWE reuses stale updates aggressively, assuming that old client innovations remain useful. In generative learning, this assumption can fail because the latent distribution changes throughout training.

To reduce harmful stale replay, we introduce Quality Echo filtering. For an unavailable client i, the echoed update is weighted as

$$w_i^t = \exp(-\lambda s_i^t) \cdot \max(0, \cos(\delta_i, \bar{\delta}^t)), \tag{3}$$

where s_i^t is the staleness of the client, δ_i is the stored echo update, and $\bar{\delta}^t$ is the average fresh update direction in the current round. The exponential term reduces the influence of old updates, while the cosine term suppresses echoes that point in a direction inconsistent with current fresh updates.

5.2. Selective Echo FedAvg-VAE

After the midterm feedback, we observed that FedAvg-VAE performed better than FedAWE-VAE in the generative setting. Therefore, instead of starting from FedAWE and trying to repair it, we designed a method that starts from FedAvg and selectively incorporates only useful FedAWE-style echoes.

Selective Echo FedAvg-VAE uses FedAvg as the primary optimization mechanism. Echoed updates are added only when they satisfy two conditions:

- the update is not too stale;
- the update direction is compatible with fresh client updates.

This makes the method more conservative than FedAWE. It avoids blindly replaying outdated updates while still allowing temporarily unavailable clients to contribute useful information.

6. Experimental Setup

We evaluate generative performance using total VAE loss, reconstruction loss, KL divergence, diversity score, generated samples, and reconstruction grids. This is important because loss alone is not sufficient to evaluate generative models.

Table 2. Experimental setup for the generative extension.

Component	Setting
Dataset	MNIST
Model	Fully-connected VAE
Latent dimension	20
Number of clients	20
Data split	Dirichlet $\alpha = 0.1$
Rounds	100
Local epochs	1
Availability	high dropout
Optimizer	Adam
Evaluation	loss, recon, KL, diversity, samples

Table 3. Performance comparison under high client dropout on MNIST with Dirichlet $\alpha = 0.1$. Lower loss and reconstruction values are better, while higher diversity is better.

Method	Loss ↓	Recon ↓	KL	Diversity ↑
FedAvg-VAE	**110.43**	**85.82**	24.61	**0.1284**
FedAWE-VAE	172.73	155.07	17.65	0.0820
Quality Echo	126.28	106.12	20.15	0.1032
Selective Echo	119.25	96.47	22.78	0.1133

7. Results

7.1. Main Quantitative Results

The results show that vanilla FedAWE-VAE performs worst. Its loss is much higher than FedAvg-VAE, and its diversity score is significantly lower. This indicates that unrestricted echo replay destabilizes generative training.

Quality Echo improves FedAWE substantially. The loss decreases from 172.73 to 126.28, showing that filtering stale and directionally inconsistent echoes reduces instability.

Selective Echo FedAvg-VAE performs best among the proposed extensions. It does not outperform FedAvg, but it comes much closer than FedAWE and preserves a higher diversity score than Quality Echo. This supports the idea that echo mechanisms are useful only when applied selectively.

7.2. Training Dynamics

Figure 1 shows the training dynamics. FedAvg-VAE converges smoothly. FedAWE-VAE shows a sharp instability spike around the early rounds and remains unstable compared to FedAvg. Quality Echo and Selective Echo both reduce this instability, with Selective Echo producing the smoothest behavior among the echo-based methods.

7.3. Generative Diversity

Figure 2 shows that FedAWE-VAE has the weakest diversity. This suggests that stale echo replay harms the learned latent

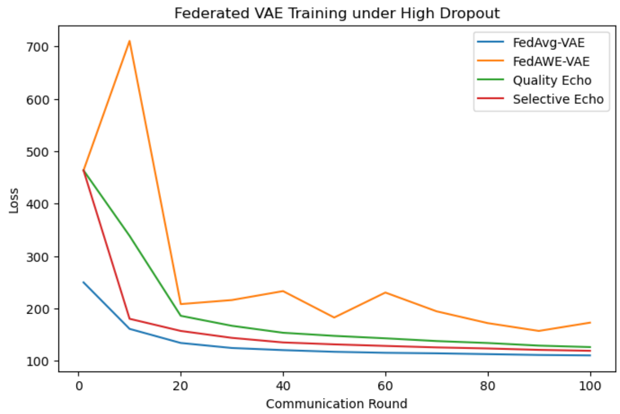


Figure 1. Loss curves under high client dropout. Vanilla FedAWE-VAE is unstable and shows a large spike early in training. Quality Echo and Selective Echo reduce this instability.

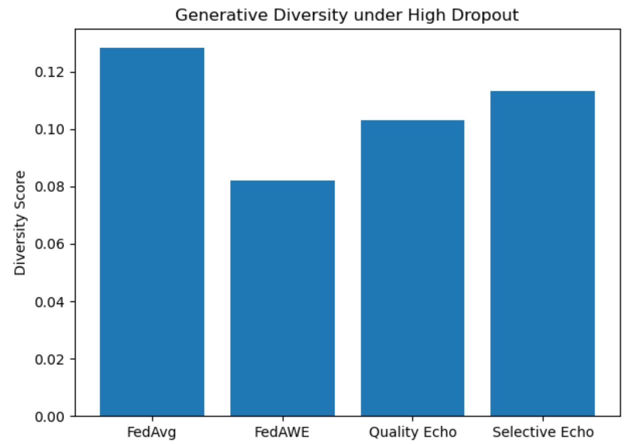


Figure 2. Diversity score comparison. FedAWE-VAE has the lowest diversity, while Selective Echo recovers much of the diversity lost by vanilla FedAWE.

distribution. Selective Echo improves diversity and moves closer to FedAvg-VAE.

7.4. Generated Samples

The generated samples provide qualitative evidence supporting the quantitative metrics. FedAvg-VAE produces the cleanest and most diverse digits. FedAWE-VAE produces blurrier and less consistent samples. Quality Echo improves sample coherence, while Selective Echo produces the strongest visual quality among the proposed echo-based variants.

7.5. Reconstruction Quality

Reconstruction quality further confirms the same trend. FedAWE-VAE loses digit structure more often than the other methods. Selective Echo preserves clearer digit shapes, supporting the conclusion that controlled echo reuse is safer than unrestricted echo replay.

7.6. Additional Validation on FashionMNIST

To verify that the observed behavior was not specific to MNIST, we additionally evaluated the methods on FashionMNIST under the same high-dropout federated setting. FashionMNIST is more challenging because it contains more complex grayscale object categories compared to handwritten digits.

The results follow the same trend observed on MNIST. FedAvg-VAE remains the most stable method, while vanilla FedAWE-VAE suffers from significant degradation under severe client dropout. Selective Echo substantially improves over vanilla FedAWE and moves closer to the FedAvg baseline.

Table 4. FashionMNIST results under high client dropout.

Method	Loss ↓	Recon ↓	KL
FedAvg-VAE	**255.22**	**240.59**	14.63
FedAWE-VAE	366.70	348.56	18.14
Selective Echo	270.30	255.32	14.97

These results suggest that the instability caused by stale replay in generative federated learning is not limited to MNIST, but also appears on more complex datasets.

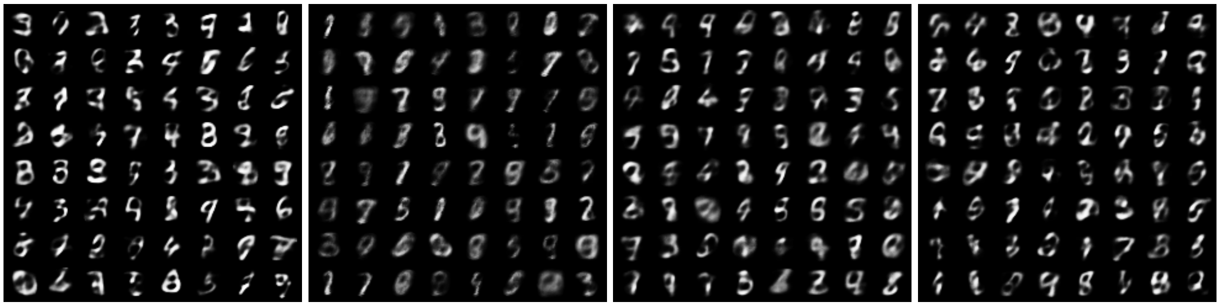
7.7. Latent Space Structure under High Dropout

To better understand the effect of stale replay on generative learning, we visualized the latent representations learned by each method using PCA projection of the VAE latent means. Figure 5 shows the structure of the learned latent space under severe client dropout.

FedAvg-VAE produces relatively coherent latent organization, where several digit classes form partially separated regions. In contrast, FedAWE-VAE exhibits significantly noisier latent representations with stronger overlap between classes. This suggests that stale replay updates distort the learned latent distribution and destabilize the representation space.

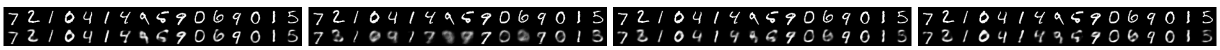
Selective Echo improves the latent structure compared to vanilla FedAWE by filtering stale updates based on staleness and directional similarity. Although the latent space remains less organized than FedAvg-VAE, the representations become noticeably more stable and clustered than those produced by FedAWE-VAE.

These observations support the hypothesis that stale replay



(a) FedAvg-VAE (b) FedAWE-VAE (c) Quality Echo (d) Selective Echo

Figure 3. Generated samples from the four methods. FedAWE-VAE produces blurrier and less diverse samples, while Selective Echo produces visually more stable samples closer to FedAvg-VAE.



(a) FedAvg-VAE (b) FedAWE-VAE (c) Quality Echo (d) Selective Echo

Figure 4. Reconstruction comparison. Top rows show original digits and bottom rows show reconstructions. FedAWE-VAE reconstructions are more degraded, while Selective Echo preserves digit structure more effectively.

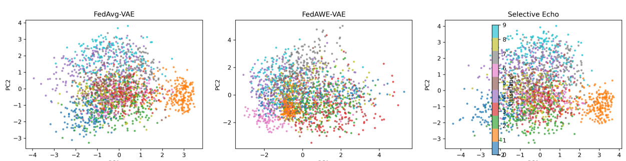


Figure 5. PCA visualization of latent representations learned by different federated VAE methods under high client dropout. Each color corresponds to a digit class. FedAvg-VAE preserves more structured latent organization, while FedAWE-VAE produces noisier and less separated latent distributions. Selective Echo partially restores latent consistency and improves stability compared to vanilla FedAWE.

is considerably more problematic in generative learning than in classification tasks, because VAEs continuously adapt a shared latent distribution rather than static decision boundaries.

8. Ablation Study

To better understand Selective Echo, we varied the maximum allowed staleness. This controls how old an echoed update is allowed to be before it is discarded.

The ablation shows that overly strict filtering removes useful client information. With max staleness 1, the method rejects many potentially useful echoes. Allowing echoes up to 3 or 5 stale rounds improves loss and diversity. In our experiment, max staleness 5 performs best, suggesting that controlled stale reuse can be beneficial when combined with filtering.

Table 5. Ablation study on maximum staleness for Selective Echo FedAvg-VAE.

Max Staleness	Loss ↓	Recon ↓	KL	Diversity ↑
1	120.69	98.26	22.43	0.1114
3	119.25	96.47	22.78	0.1133
5	**119.04**	**96.22**	22.82	**0.1155**

9. Discussion

The main finding of this project is that FedAWE does not transfer directly from federated classification to federated generative learning. In the original classification setting, stale updates may still contain useful decision-boundary information because decision boundaries typically evolve relatively smoothly during training. However, Variational Autoencoders (VAEs) learn a continuously evolving latent distribution, making them substantially more sensitive to stale or inconsistent updates.

Under severe client dropout, stale echoed updates may correspond to outdated latent-space directions. Replaying these stale updates aggressively can distort the global latent representation and destabilize generative optimization. This explains why vanilla FedAWE-VAE performs significantly worse than FedAvg-VAE in our experiments.

FedAvg only aggregates fresh updates from currently available clients, which makes optimization more stable for VAE training. However, completely ignoring unavailable clients also discards potentially useful information. To address this trade-off, we proposed two stabilization mechanisms: Quality Echo and Selective Echo.

Quality Echo reduces the influence of stale replay by weighting echoed updates according to their staleness and directional consistency with fresh updates. This substantially improves over vanilla FedAWE-VAE and demonstrates that replay filtering can partially stabilize generative training.

After observing that FedAvg-VAE consistently outperformed FedAWE-VAE, we further designed Selective Echo FedAvg-VAE. Instead of relying primarily on stale replay, this method keeps FedAvg as the stable optimization backbone and selectively reuses only safe stale updates. Experimental results show that Selective Echo achieves substantially lower loss and higher diversity than vanilla FedAWE while remaining significantly more stable under severe client dropout.

Table 6 summarizes the behavioral difference between the original FedAWE classification setting and our federated generative VAE extension. In the original paper, FedAWE improves robustness under client unavailability. In contrast, our experiments reveal that unrestricted stale replay becomes harmful in federated generative learning because outdated latent-space information destabilizes VAE optimization.

To verify that the observed instability was not limited to MNIST, we additionally evaluated the methods on FashionMNIST under the same high-dropout federated setting. The same behavioral trend appears on this more challenging dataset: FedAvg-VAE remains the most stable method, vanilla FedAWE-VAE suffers from severe degradation, and Selective Echo substantially improves over vanilla FedAWE.

9.1. Why FedAWE Becomes Unstable in Generative Learning

Unlike classification models, Variational Autoencoders (VAEs) optimize a continuously evolving latent distribution through both reconstruction and KL regularization. As a result, generative learning is substantially more sensitive to stale or inconsistent updates.

FedAWE assumes that previously observed client updates remain useful after temporary client unavailability. While this assumption is often effective for classification tasks, stale replay becomes problematic for VAEs because replayed updates may correspond to outdated latent-space structures.

Under severe client dropout and highly non-IID partitions, stale updates inject inconsistent optimization directions into the latent space, destabilizing both reconstruction quality and latent regularization. This explains the significantly higher loss and lower diversity observed in vanilla FedAWE-VAE.

Our experiments suggest that replay mechanisms designed for discriminative learning cannot be transferred directly to federated generative learning without additional stability constraints. This observation motivated the development of Quality Echo and Selective Echo stabilization mechanisms.

9.2. Client Availability Scenarios

The original FedAWE paper primarily evaluates periodic client availability fluctuations using sine-based participation patterns. In this work, we additionally introduce more challenging availability scenarios specifically designed for federated generative learning.

The evaluated settings include:

- **Stationary:** clients participate with relatively stable probability across communication rounds.
- **Sudden Drop:** client availability decreases sharply after a certain training stage, simulating unexpected large-scale client disconnection.
- **High Dropout:** only a small subset of clients is available at each round, representing highly sparse participation conditions.

These scenarios were intentionally selected to stress-test stale replay mechanisms under severe non-stationary participation. Generative models are substantially more sensitive to stale latent-space updates than classification models, making such challenging availability settings particularly important for evaluating training stability.

10. Limitations and Future Work

This paper has several limitations. First, the generative experiments are conducted on MNIST using a relatively small fully-connected VAE. Larger datasets and more complex generative architectures may show different behavior. Second, although we add diversity and visual evaluation, we do not compute FID, which would provide a stronger sample-quality metric. Third, we do not study adversarial clients, where malicious clients may intentionally exploit the echo mechanism. Finally, we focus on image generation only and do not explore multimodal or text-based generative models.

Future work should evaluate Selective Echo on Fashion-MNIST, CIFAR-based generative models, GANs, and diffusion models. Another important direction is adversarial unavailability, where clients strategically go offline after submitting harmful updates. This would connect the current

Table 6. Behavioral comparison between the original FedAWE classification setting and our federated generative VAE extension. Classification results are reported as test accuracy, while generative learning results are reported using VAE loss and diversity metrics.

Setting	Method	Final Loss	Recon Loss	KL Loss	Diversity	Observation
Original Paper (CIFAR-10 Classification)	FedAvg	62.9% accuracy	–	–	–	Lower robustness under unavailability
Original Paper (CIFAR-10 Classification)	FedAWE	66.3% accuracy	–	–	–	Improves over FedAvg under dynamic availability
Our Work (MNIST VAE)	FedAvg-VAE	110.43	85.82	24.61	0.128	Stable convergence and best generative quality
Our Work (MNIST VAE)	FedAWE-VAE	172.73	155.07	17.65	0.082	Severe instability caused by stale replay
Our Work (MNIST VAE)	Quality Echo	126.28	106.12	20.15	0.103	Partial stabilization using weighted replay
Our Work (MNIST VAE)	Selective Echo	119.04	96.22	22.82	0.116	Improved stability via controlled stale replay

Table 7. Main quantitative results under high client dropout on MNIST. Lower loss is better. Higher diversity is better.

Method	Loss ↓	Recon ↓	KL	Diversity ↑
FedAvg-VAE	**110.43**	**85.82**	24.61	**0.128**
FedAWE-VAE	172.73	155.07	17.65	0.082
Quality Echo	126.28	106.12	20.15	0.103
Selective Echo	119.04	96.22	22.82	0.116

Table 8. FashionMNIST validation under high dropout (50 rounds).

Method	Final Loss	Observation
FedAvg-VAE	255.22	Most stable convergence
FedAWE-VAE	366.70	Strong instability under stale replay
Selective Echo	270.30	Significant recovery over FedAWE

project to robustness and security in federated generative learning.

11. Conclusion

We reproduced the FedAWE setting and extended it from federated classification to federated generative learning. Our experiments show that vanilla FedAWE becomes unstable for VAE training under severe client dropout because unrestricted stale replay distorts the learned latent distribution. To address this limitation, we proposed Quality Echo filtering and Selective Echo FedAvg-VAE. Results across MNIST, FashionMNIST, generated samples, reconstruction analysis, latent-space PCA, and ablation studies show that selective replay stabilization substantially reduces FedAWE instability while preserving generative quality. Overall, our findings suggest that replay mechanisms designed for discriminative federated learning cannot be transferred directly to generative federated optimization without additional stability constraints.

Software and Data

The implementation includes scripts for FedAvg-VAE, FedAWE-VAE, Quality Echo, Selective Echo, evaluation metrics, generated samples, reconstruction grids, and plotting. The repository contains the experimental CSV files, trained checkpoints, and visual results used in this report.

Impact Statement

This project studies federated learning under client unavailability and aims to improve the robustness of distributed machine learning systems. The work may contribute to privacy-preserving learning in practical environments where client participation is unreliable. Since the project uses MNIST and simulated client dropout, it does not directly involve sensitive real-world user data. However, future applications of federated generative learning should carefully consider privacy, fairness, and potential misuse of synthetic data generation.